GG-B-675E
June 28, 1988
SUPERSEDING
GG-B-675D
September 23, 1976
(See section 6)

FEDERAL SPECIFICATION

BREATHING APPARATUS, SELF-CONTAINED

This specification is approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 Scope. This specification covers self-contained pressure demand type breathing apparatus with audible warning device for respiratory protection in oxygen-deficient or toxic atmospheres containing vapors or gases. Protection is not provided against poisoning by skin absorption.

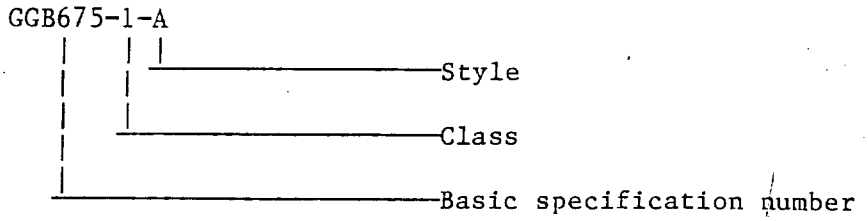
1.1.1 Federal specification coverage. Federal specifications do not include all varieties of the commodity as indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification (see 6.3).

1.2.1 Class and style. Self-contained breathing apparatus furnished in accordance with this specification shall be the one-half (1/2) hour duration open circuit design. They shall be of the following classes and styles as specified (see 6.2):

Class 1 - Compressed air
Class 2 - Compressed oxygen
Style A - Waist strap mounted regulator
Style B - Face-mounted regulator

1.2.2 Definitive part number (DPN). The breathing apparatus shall be identified by a DPN as shown in the following example:



FSC 4240

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2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

Federal Specifications:

PPP-B-601 - Boxes, Wood, Cleated-Plywood
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner
PPP-B-636 - Box, Shipping, Fiberboard
PPP-C-795 - Cushioning Material, Packaging (Flexible Cellular, Plastic Film) For Packaging Applications
PPP-C-843 - Cushioning Material, Cellulosic
PPP-F-320 - Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes

Federal Standard:

FED. TEST METHOD STD. NO. 191 - Textile Test Methods

(Activities outside the Federal Government may obtain copies of Federal specifications and standards as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Washington, DC; Philadelphia, PA; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specification:

MIL-P-116 - Preservation, Methods of

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

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(Copies of military specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

Code of Federal Regulations (CFR):

29 CFR 1910 - OSHA Safety and Health Standards
30 CFR 11 - Respiratory Protective Devices; Tests for Permissibility; Fees
49 CFR 170-179 - Transportation

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

Department of the Army Technical Manual:

TM38-250 - Preparation of Hazardous Materials for Military Air Shipments

(Copies are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Approval. Self-contained breathing apparatus furnished under this specification shall comply with the requirements specified in 30 CFR 11 and 29 CFR 1910 (see 4.2).

3.1.1 Plates and markings. The breathing apparatus shall bear approval plates and markings for permissable apparatus and for approved supplemental parts as provided in 30 CFR 11 subpart D and 29 CFR 1910.134(d)(4).

3.2 Compliance. Compliance with the above requirements does not absolve the bidder from complete compliance with the requirements of this specification in order to secure acceptance of his product.

3.3 Design. The apparatus shall be an individual respiratory protective device designed to provide the user with a respirable atmosphere from compressed gas cylinder(s). The apparatus shall consist of three major groups: (1) the facepiece group, (2) the hose and regulator group, and (3) the backpack and harness group. A suitcase type carrying case (see 3.5.8) shall be furnished with each apparatus for transportation and storage purposes.

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3.4 Material. All apparatus furnished under this specification shall be of the manufacturer's most recent production model approved in accordance with 30 CFR 11 subpart D. The materials used for the manufacture and assembly of the apparatus shall be new.

3.5 Components.

3.5.1 Interchangeability. Any manufacturer's (contractor's) components used in the assembly of the breathing apparatus shall be interchangeable with others of the same component furnished by that manufacturer.

3.5.2 Facepiece. The facepiece shall be of the design that covers the entire face and shall be so constructed as to assure a quick gas-tight fit on persons of varying facial shapes. A means shall be provided so that the facepiece may be suspended on the user in a convenient-to-use location when the apparatus is donned but not in use.

3.5.2.1 Lenses. The facepiece shall be equipped with a single lens. Lens replacement shall be readily accomplished in the field.

3.5.3 Compressed gas cylinders. Compressed gas cylinders shall conform to 30 CFR 11 subpart H.

3.5.4 Harness assembly. The apparatus shall be provided with an adjustable fit harness for comfortably supporting the apparatus securely on the wearer's back. The harness shall include suitable means to prevent shifting of the apparatus on the wearer's back, during physical work. The shoulder strap or connection shall have a breaking strength of not less than 175 pounds (see 4.4.1).

3.5.5 Audible warning device. The breathing apparatus shall be equipped with a warning device. The warning device shall sound an audible alarm when the remaiing service life of the apparatus is reduced within a range of 20 to 25 percent of its rated service time.

3.5.6 High pressure hose leakage. The high pressure hose connecting the breathing air cylinder to the regulator shall not leak when tested as specified in 4.4.3.

3.5.7 Breathing tube assembly (style A only). The breathing tube assembly shall be a corrugated flexible hose molded from an elastomer or other nonmetallic material. Wire or wire braid reinforced hose is acceptable. The length shall be sufficient to connect the facepiece to the regulator so that when worn, free movement of the head in any direction will be permitted without kinking the hose or placing stress on the hose connections.

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3.5.8 Carrying case.

3.5.8.1 Design and construction. A carrying case shall be provided to retain the breathing apparatus, manuals, and wrench, as specified (see 6.2). The case may be made of molded plastic, aluminum alloy, vulcanized fiber, or other suitable material. When aluminum alloy is used, it shall be anodized or treated with a chromate chemical conversion process and then shall be coated with one coat of zinc chromate primer and one coat of enamel. The case shall be provided with corrosion-resistant hinges and catches to insure closure, and a handle for ease of carrying. The case shall be of a design to provide compact enclosure of the apparatus and accessories without danger of deformation of any part due to closure. The case shall be provided with suitable built-in brackets, bracing, or straps for holding the apparatus securely in place under severe handling conditions. Sponge rubber or other suitable cushioning material shall be provided at points where shock stresses could be transmitted to the apparatus. The weight of the carrying case shall not exceed 25 pounds (see 5.2).

3.5.8.2 Manuals. One copy each of the manufacturer's standard operator and maintenance manual and repair parts list shall be furnished with each apparatus.

3.5.8.3 Wrench. When required by the manufacturer's design, a suitable wrench for connecting the high pressure hose to the compressed-gas cylinder and the regulator shall be furnished with each apparatus.

3.6 Workmanship. The apparatus shall be free from contamination (grease, oil, or other foreign matter) and damage (dents, scratches, or punctures) and other defects which might affect proper functioning.

3.6.1 Breathing tubes (style A only). The breathing tubes shall be free from tackiness, porous areas, crack, flashes, abrasions, and blisters. The elastomer shall be compact and free from bubbles and foreign matter. Ridges of mold parting lines, when closely trimmed, shall not be considered as flash.

3.6.2 Metal surfaces. The metal parts shall have no exterior sharp edges or burrs.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Certification. The contractor shall certify in writing to the contracting officer or his representative that the items he proposes to furnish the Government under this specification are jointly approved by the National Institute for Occupational Safety and Health (NIOSH) and the Mine Safety and Health Administration (MSHA), U.S. Department of Labor, under 30 CFR 11 subpart H. The certification shall include the NIOSH approval number.

4.3 Sampling. Two samples of each class breathing apparatus assemblies shall be selected at random from the first lot produced for the tests specified in 4.4.1 through 4.4.3. If a sample fails to comply with any of these tests, two additional samples shall be selected. If any additional sample fails any of the tests, breathing apparatus represented by the samples shall be considered as not conforming to the requirements of this specification. Manufacturers may correct minor deficiencies that cause failure. Lots subsequent to the first lot shall be tested only upon change in design, material, method of manufacture, or location of manufacturing facility. Sampling for visual examination (see 4.4.4) shall be 100 percent.

4.4 Test methods.

4.4.1 Strength of shoulder straps. The test shall be conducted in accordance with method 4108 or 5100 of FED. TEST METHOD STD. NO. 191. When method 5100 is used, the distance between the clamps shall be 6 inches at the start of the test. All joints under test shall be midway between the clamps in the plane of pull.

4.4.2 Audible alarm operation. The cylinder shall be charged to 40 percent or greater of its rated pressure. The cylinder valve shall be fully opened to pressurize the entire system. The by-pass valve shall be opened slowly to vent the system pressure, observing the pressure turn-on of the audible alarm. The pressure at which the alarm starts to sound and ceases to sound shall be recorded, using a master gage of sufficient sensitivity. The pressure readings shall be within the limits specified in 3.5.5.

4.4.3 High pressure hose test.

4.4.3.1 Style A. Connect the high pressure hose to a fully charged breathing air cylinder and regulator in accordance with the manufacturer's instructions. Close the regulator shut-off and by-pass valves and fully open the cylinder valve. Apply soapy water to the hose and observe for conformance to 3.5.6. The appearance of bubbles indicates a leak.

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4.4.3.2 Style B. The mask mounted regulator is to be suitably capped and the purge valve fully closed. Then connect the pressure reducer high pressure hose to the cylinder valve, open the cylinder valve and leak check. Apply soapy water to the hose and observe for conformance to 3.5.6. The appearance of continuous bubbles indicates a leak.

4.4.4 Visual examination. The apparatus shall be visually examined for compliance with the requirements of 3.1, 3.5, and 3.6.

4.5 Inspection of packaging requirements. An inspection shall be made to determine whether the packaging, packing and marking requirements comply with section 5 of this specification. Defects shall be scored in accordance with table I. The sample unit shall be one shipping container fully prepared for delivery. Sampling shall be in accordance with MIL-STD-105. The lot sizes shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 3.5 defects per hundred units.

TABLE I. Classification of packaging defects

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Distortion of container.

5. PACKAGING

5.1 Cleaning and drying. Prior to preservation and unit packing, the breathing apparatus and components shall be cleaned and dried in accordance with cleaning method C-1 of MIL-P-116 and any of the drying methods listed therein.

5.2 Preservation and unit packing, level A. The apparatus with components, assembled in the carrying case as specified in 3.5.8, shall be cushioned as required to prevent movement using cushioning material conforming to PPP-C-795 or PPP-C-843. The carrying case shall be closed with fasteners provided. The apparatus shall be unit packed in a snug fitting fiberboard box conforming to type RSC, class WR, grade W5c of PPP-B-636. Filler pads conforming to the same material as the fiberboard box shall be used as necessary to make a tight assembly. The box shall be closed in accordance with the method IV closure requirements in the appendix to PPP-B-636.

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5.3 Packing. Packing shall be level A or B as specified (see 6.2).

5.3.1 Level A. The breathing apparatus, unit packed as specified in 5.2, shall be packed in a snug fitting overseas wood box conforming to style optional, class 2, grade A, type 1 load, 0-50 pound load limit, of PPP-B-621. As an alternative, a cleated-plywood box conforming to overseas type, style optional, grade A, type 1 load, 0-100 pound load limit, of PPP-B-601, may be used. Filler pads, fabricated of fiberboard conforming to grade W5c or W6c of PPP-F-320, shall be used as necessary to make a tight pack. The boxes shall be closed and strapped in accordance with the appendix to their respective specifications.

5.3.2 Level B. The breathing apparatus, unit packed as specified in 5.2, shall serve as the level B pack, except that the box shall be closed in accordance with the method V closure requirement and reinforced in accordance with the appendix to PPP-B-636.

5.4 Marking.

5.4.1 Unit packing and packing. In addition to any special markings required in the contract or order (see 6.2), the unit packs (see 5.2) and packs (see 5.3) shall be marked in accordance with MIL-STD-129.

5.4.2 Special marking. Shipments shall be marked in accordance with 49 CFR 170-179 and TM38-250 as follows:

(a) Proper shipping name: Air, compressed, UN1002.

(b) DOT label: Non-flammable gas, with a "2" in bottom corner of label.

6. NOTES

6.1 Intended use. Breathing apparatus in accordance with this specification are designed for emergency entrance into a hazardous atmosphere. They are not for underground mine rescue or recovery operations except as auxiliary equipment.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification
(b) Class and style required (see 1.2.1)
(c) Level of packing required (see 5.3)
(d) Special marking, if required (see 5.4)
(e) Wrench, if required (see 3.5.8)

6.3 Classification of previous issue. The following classification of breathing apparatus was contained in the previous issue of this specification:

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Class 1 - Compressed air
Class 2 - Compressed oxygen
Style B - Demand type with audible warning device
Style D - Pressure demand type with audible warning device

The styles have been redefined in this revision because the demand type breathing apparatus (style B of previous issue) is no longer recognized by NIOSH as being acceptable for use. Styles A and B of this specification represent pressure demand type apparatus (style D of previous issue.).

6.4 Supersession data. This specification also supersedes and includes the requirements of Interim Amendment-3 (Army/EA), dated Spetember 9, 1986.

MILITARY INTERESTS:

Custodian:

Army - EA

Review activities:

Army - MD

CIVIL AGENCY COORDINATING ACTIVITY:

GSA-7FXEE

Preparing activity:

Army - EA

Project No. 4240-0545

Orders for this publication are to be placed with the General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain copies and other documents referenced herein.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
GG-B-675B	BREATHING APPARATUS, SELF-CONTAINED

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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